

B. TECHNICAL REQUIREMENTS SPECIFICATION

5. SYSTEM ANALYSIS

5.1 Current System Analysis

The university clinic of National University Dasmariñas currently relies on a manual, paper based process for recording, scheduling, and tracking patient visits. Close to one hundred people use the clinic on an average day, including college students, senior high school students, faculty, staff, employees, and outside visitors. The clinic follows a triage system, where urgent cases are attended to before non urgent ones, and every visit goes through the same series of steps.

First, the nurse asks the patient for the purpose of the visit and decides how urgent it is. Cases such as accidents, bleeding, and fainting are attended to right away, while less urgent cases such as dizziness, coughing, or a clearance request are put on hold. When the clinic becomes crowded, these patients may be asked to wait outside. Second, the nurse retrieves the patient's physical folder from the filing cabinet, which is arranged alphabetically by academic program. Third, the nurse takes the patient's vital signs by hand and writes them directly on the paper record. Fourth, the record is handed to the doctor on duty, who then calls the patient in for consultation.

Fifth, based on the doctor's assessment, the patient may be given medicine, allowed to rest in a clinic bed, sent back to class, sent home, or referred to a hospital if the case goes beyond what the clinic can handle as a first aid station. For minor ailments, the stay in a clinic bed is limited to 30 minutes to 1 hour. If the patient still does not feel well after that time, a handwritten medical certificate is issued and the patient is sent home, while a minor must be picked up by a parent. Sixth, once the visit ends, the nurse files the record back into the cabinet. Any medicine given during the visit is written in the clinic logbook and counted by hand at the end of the day. The nurses then encode these entries into an Excel file, which is used to track the remaining stock and later to prepare the clinic's monthly report.

Doctor and dentist schedules are posted only on the clinic's Facebook page, and appointments are taken through a Microsoft Forms link that does not show a live queue. This setup creates several problems. Searching through filing cabinets is slow because the university has roughly 9,000 students, and misplaced folders often lead to duplicate records for the same patient. Outside visitors have no folder at all, so there is no consistent way to record them. The accuracy of every record depends on the nurses copying each entry correctly from the logbook, to the Excel file, and to the patient folder, which takes time away from patient care and leaves room for human error. Patients cannot check the doctor's schedule ahead of time, and during busy periods such as enrollment, the lack of a visible queue leads to long lines and crowding outside the clinic. Paper files also give no control over who can read or remove a record.

The clinic's daily work, scheduling, and inventory are also kept separate from one another, with no central system tying them together. The university has its own centralized system called NUIS, short for National University Information System, which gives students and staff general resources and communication updates. The clinic does not connect to NUIS in any way, so its data stays outside the university's digital system even though that infrastructure already exists

5.2 Proposed System Analysis

MediSync is designed to address the problems of the current process by replacing the paper based workflow with one centralized digital system. It will improve the record keeping process by storing every patient record in a database instead of logbooks and filing cabinets, so records are easier to search and misplaced folders and duplicate files are prevented. Enrolled students and personnel will be checked in through RFID tap in, while outside visitors will be recorded through a guided registration, so everyone the clinic serves has a record. Scheduling will be improved through online appointment booking, a live display of doctor and dentist schedules, and a digital priority queue that sorts each visit by urgency. Inventory tracking and monthly reporting will also be improved, since stock levels and consultation data will be kept in the system instead of in separate logbooks and Excel files.

Many tasks that are done by hand today will be automated or simplified. Medicine stock will be deducted automatically whenever a medicine is dispensed, and the system will send low stock alerts so shortages are noticed early. Vital signs and consultation details will be entered into digital forms based on the clinic's existing forms, and medical certificates will be generated from the consultation record using digital templates, which staff review before release. Monthly reports will be generated from stored data, which removes the need to copy figures into a separate Excel file. Appointment confirmations and clinic announcements will be sent automatically through in app notifications, SMS, and the school Outlook email. The system will also suggest a priority level for each visit, which clinic personnel can review and adjust, so the final decision on priority always stays with the staff.

Students, faculty, and staff will use the MediSync mobile application. They will log in, accept the terms and conditions consent, book appointments, view doctor and dentist schedules, track their queue number, and receive reminders. Clinic personnel will use the web portal with role based access control, so a nurse, doctor, dentist, or administrator only sees the information needed for their role. Through the portal, they will manage patient records, confirm the queue, manage the medicine inventory, fill out consultation forms, issue prescriptions and certificates, and view reports. Basic student details will be taken from the university's existing student information system, which lessens repeated encoding. Emergency cases will still be handled in person by clinic staff and are outside the scope of the system.



The proposed system differs from the current process in how information is stored, accessed, and shared. The current process depends on handwritten records, locked filing cabinets, handwritten certificates, and manual counting, while the proposed system keeps this information in one secure database that authorized users can search in seconds. In the current process, patients walk in without knowing the schedule or how long the wait will be, but with MediSync they can book ahead, see the schedule, and follow their place in the queue, which helps prevent overcrowding. Inventory and monthly reports are no longer prepared through separate logbooks and spreadsheets but are produced automatically by the system. Access to records is also controlled by role, which gives the clinic a safer way to handle sensitive health information than a paper file can.



5.3 Process Model

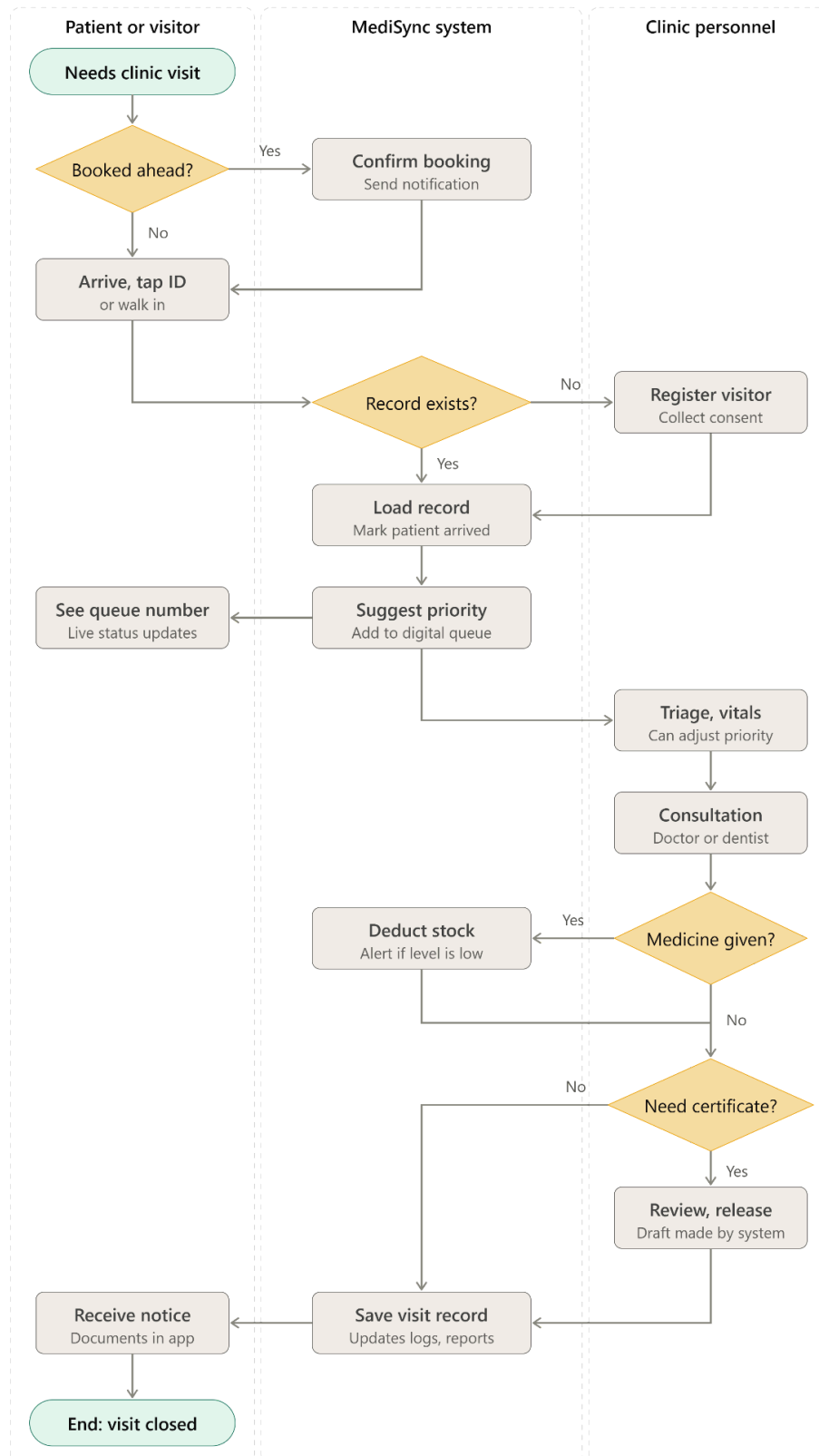


Figure 5.3.1 Process Model of a Clinic Visit in MediSync

This process model is a swimlane flowchart that follows one clinic visit from start to end across three parties, which are the patient or visitor, the MediSync system, and the clinic personnel. The flow begins when a person needs a clinic visit and ends when the visit is closed and the person receives a notice. Rounded shapes mark the start and end points, rectangles mark the major activities, and diamonds mark the decisions. Arrows show how actions and information move from one party to the next.

The first decision asks whether the person booked ahead. If yes, the system confirms the booking and sends a notification, and the person later arrives at the clinic and taps the school ID. If no, the person simply arrives and taps the ID or walks in. The second decision checks whether a record already exists. Enrolled students and personnel already have a record, so the system loads it and marks the patient as arrived. Outside visitors do not have one, so clinic personnel register them and collect their consent before the visit continues.

After check in, the system suggests a priority level and adds the patient to the digital queue, where the patient can see the queue number. A nurse then does the triage, takes the vital signs, and may adjust the suggested priority. The doctor or dentist then holds the consultation. The third decision asks whether medicine was given. If yes, the system deducts the stock and sends an alert when the level is low. The fourth decision asks whether a certificate is needed. If yes, the system prepares a draft that clinic personnel review and release. If no, the process goes straight to the last step. Both paths end with the system saving the visit record, which also updates the logs and monthly reports, and the patient receives a notice in the app.